

KRISH AGRAWAL

Data Engineer | Backend Developer

Mathura, Uttar Pradesh | krishagrawal706krish07@gmail.com | github.com/krishagrawal07 | linkedin.com/in/krish-agrawal-0309152a8

SUMMARY

B.Tech (Computer Science & Engineering — AI/ML) undergraduate with hands-on experience building data pipelines and backend services in Python. Comfortable across the stack from stream processing (Kafka, Spark) to warehouse modeling (dbt, SQL) to API development (FastAPI, Redis), with all projects containerized and version-controlled. Seeking a Data Engineering or Backend Development role to apply strong fundamentals in distributed systems, data infrastructure, and clean software design.

TECHNICAL SKILLS

Languages: Python, SQL, Go, Java

Data Engineering: Apache Spark, Apache Airflow, Apache Kafka, dbt, ETL/ELT Pipeline Design

Databases: PostgreSQL, Snowflake, Redis, MongoDB

Backend & APIs: FastAPI, REST API Design, Pydantic, pytest

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, CI/CD, Git

PROJECTS

Realtime Clickstream Pipeline — *Python, Kafka, PySpark Structured Streaming* github.com/krishagrawal07/realtime-clickstream-pipeline

- Built an end-to-end streaming pipeline that generates simulated clickstream events and publishes them to a Kafka-compatible broker (Redpanda) using kafka-python.
- Implemented a PySpark Structured Streaming job with a 1-minute tumbling window and watermarking to compute live per-page view counts.
- Designed the aggregation layer to be swappable to a Parquet/Delta sink for downstream warehouse loads.

Warehouse Sync Service — *Python, SQLAlchemy, PostgreSQL, dbt* github.com/krishagrawal07/warehouse-sync-service

- Developed an incremental ELT service that syncs changed rows from a source database into a warehouse using updated_at watermarking and idempotent upserts.
- Built dbt staging and mart models to transform raw synced data into analytics-ready tables, with an automated data-quality test on order totals.
- Architected the connection layer with SQLAlchemy so the target can be swapped from PostgreSQL to Snowflake with no logic changes.

Internal Metrics API — *Python, FastAPI, Redis, Docker* github.com/krishagrawal07/internal-metrics-api

- Built a REST API serving aggregated product metrics with FastAPI, using Redis to cache responses (30s TTL) and cut repeated computation latency from ~500ms to under 10ms on cache hits.
- Wrote a fully offline automated test suite with pytest and fakeredis, covering health checks, caching behavior, and error handling — all tests passing.
- Containerized the API and Redis with Docker Compose for a reproducible one-command local setup.

Batch Job Orchestrator — *Python, Apache Airflow, Docker* github.com/krishagrawal07/batch-job-orchestrator

- Designed a reusable Airflow framework standardizing retry policy, exponential backoff, SLA tracking, and failure alerting across scheduled data jobs.
- Built an hourly ETL DAG (sync -> dbt run -> dbt test) and a config-driven backfill DAG supporting arbitrary date-range reprocessing.
- Reduced per-job boilerplate by centralizing orchestration logic into a shared plugin used by every DAG.

EDUCATION

GLA University, Mathura

Currently Pursuing

Bachelor of Technology (B.Tech), Computer Science & Engineering — AI/ML

CGPA: 8.4 / 10 (First Year)